

3.

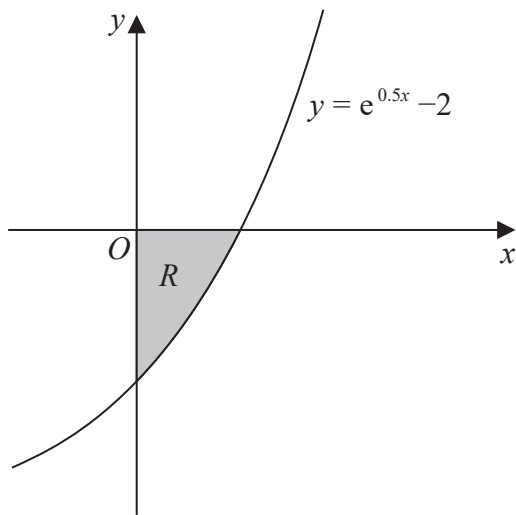
**Figure 1**

Figure 1 shows a sketch of part of the curve with equation $y = e^{0.5x} - 2$

The region R , shown shaded in Figure 1, is bounded by the curve, the x -axis and the y -axis.

The region R is rotated 360° about the x -axis to form a solid of revolution.

Show that the volume of this solid can be written in the form $a \ln 2 + b$, where a and b are constants to be found.